

ECON 0100 | Homework C1 | Tariffs

Due: TBD

Homework is designed to both test your knowledge and challenge you to apply familiar concepts in new applications. Answer clearly and completely, and show your work so you can later understand your thought process. You are welcomed and encouraged to work in groups as long as your work is your own. Submit your answers on Gradescope when you're finished: each answer there is a selection.

Butterbeer

Preferences for butter beer can be represented by the following demand curve:

$$P_d = 100 - \frac{1}{2}Q_d$$

The supply curve for butter beer can be represented by the following equation:

$$P = 20 + \frac{1}{2}Q$$

Prices are in galleons and quantity is in bottles.

Q1 | Opening the Market

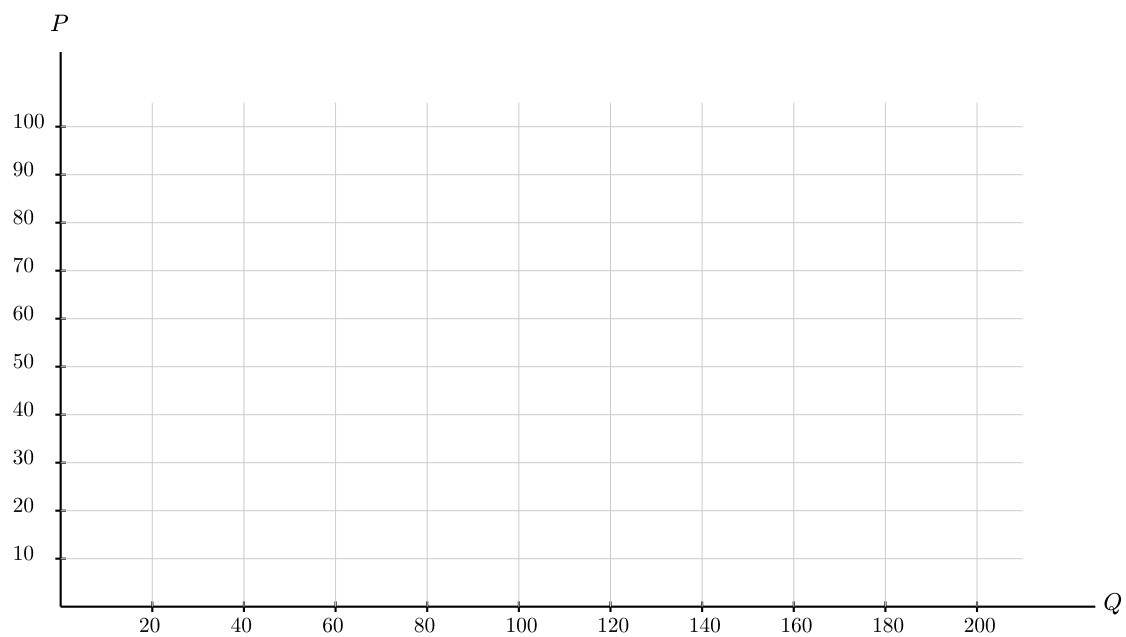
The Ministry of Magic hired an external committee of advisors. One classically trained magical economist proposed opening the closed domestic butterbeer market to the less expensive global market ($P = 40$). Ministry officials were pleased with the option but also raised concerns that domestic butterbeer sellers would be harmed by the lower price. Use a graph and algebra to analyze the impact.

- a) Imports or exports? *imports, exports*
- b) Quantity demanded: _____ Quantity supplied: _____ Imports: _____
- c) Consumer surplus before: _____ after: _____

d) Producer surplus before: _____ after: _____

e) Change in total surplus: _____

f) Buyers: *gain, lose* Sellers: *gain, lose*



Q2 | An Alternative World

If instead of a price of 40 galleons in Question 1 the global price had been 80 what would have been the distributional impact of opening the domestic market for trade? Use a graph and algebra.

a) Imports or exports? *imports, exports*

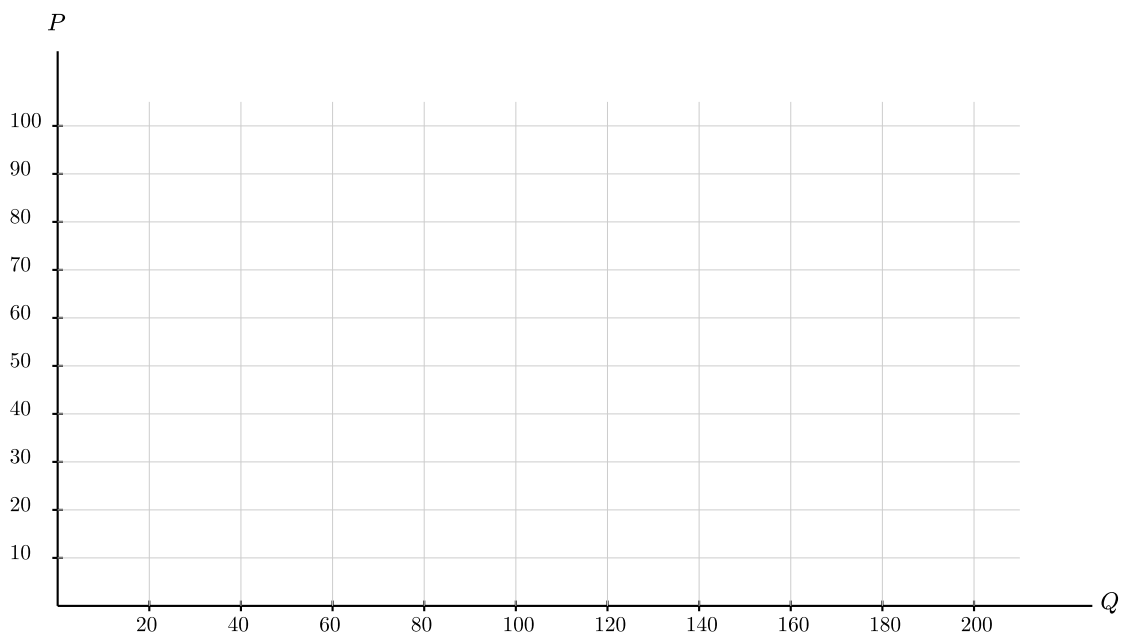
b) Quantity demanded: _____ Quantity supplied: _____ Exports: _____

c) Consumer surplus before: _____ after: _____

d) Producer surplus before: _____ after: _____

e) Change in total surplus: _____

f) Buyers: *gain, lose* Sellers: *gain, lose*



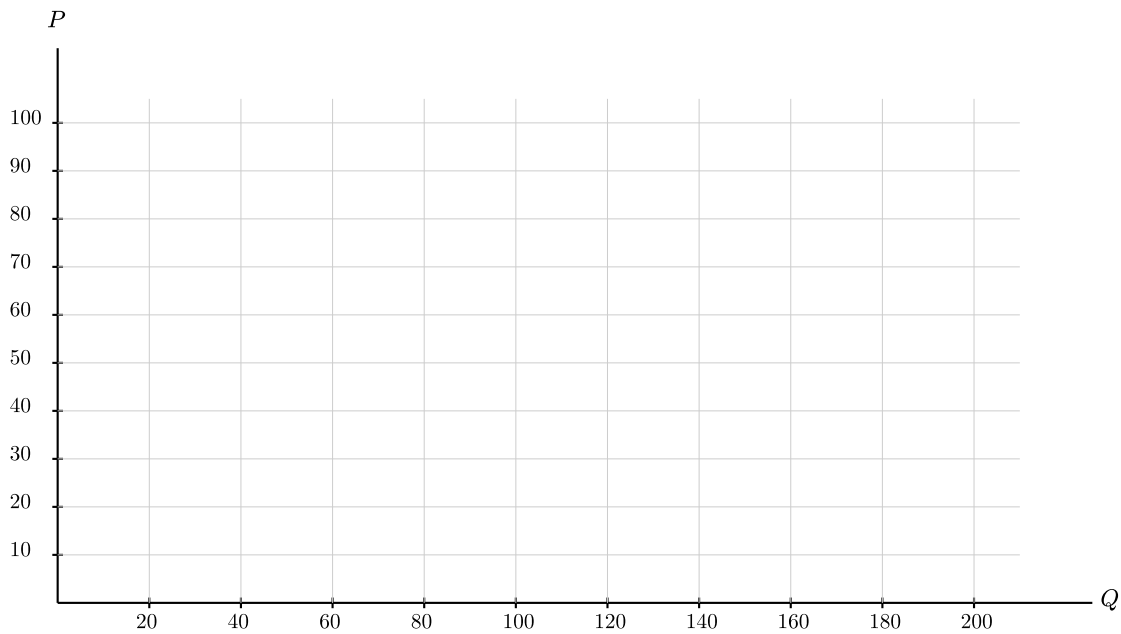
Q3 | A Changing World Price

Return to Q1. Suppose the global price rises from 40 to 50 galleons.

a) Quantity demanded: _____ Quantity supplied: _____ Imports: _____

b) Consumer surplus: _____ Producer surplus: _____

c) Imports have: *risen, stayed the same, fallen*



Q4 | A Tariff on International Butterbeer

The Ministry of Magic hired an external committee of advisors to find means by which to raise government revenue without raising taxes. One classically trained magical economist proposed opening the closed domestic butterbeer market to the less expensive global market ($P = 10$) while imposing an import tariff of 2 galleons. Ministry officials were pleased with the option but also raised concerns that domestic butterbeer sellers would be harmed by the lower price.

Use a graph to illustrate what would happen to the market after a tariff was imposed to answer the ministry's concerns. The supply and demand curves for butterbeer can be represented by the following equations.

$$P = 22 - \frac{1}{6}Q_D$$

$$P = 2 + \frac{1}{2}Q_S$$

Q5 | Political Backlash

In response to anticipated political backlash associated with raising revenue via a tax on the much beloved butterbeer, the Ministry of Magic hired an external committee of advisors to find other means by which to raise government revenue. One classically trained magical economist proposed opening the closed domestic butterbeer market to the less expensive global market while imposing an import tariff. Ministry officials were pleased with the option but also raised concerns that domestic butterbeer sellers would be harmed by the lower price.

Use a graph to illustrate what would happen to the market after a tariff was imposed. Is there a tariff price that would alleviate the Ministry's concern about domestic sellers?

The supply and demand curves for butterbeer can be represented by the following equations.

$$P = 23 - \frac{1}{6}Q$$

$$P = 2 + \frac{1}{2}Q$$